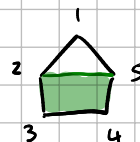


1 BP

$$\begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} - \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} = 0$$

$$\begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} = 0$$

$$\begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} = 0$$



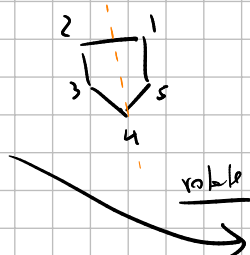
2 BP

$$\begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} - \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} + \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} = 0$$

$$\begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} = 0$$

$$\begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} - \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} - \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} = 0$$

$$\begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} + \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} - \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} = 0$$



$$- \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} - \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} + \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} + \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} = 0$$

$$- \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} + 2 \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} + \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} - \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 2 \quad 5 \\ \diagdown \quad \diagup \\ 3 \quad 4 \end{array} = 0$$

$$A_S(1, 2, 3, 4, 5) = \frac{a_{13,14}}{x_{13} x_{14}} + \frac{a_{13,35}}{x_{13} x_{35}} + \frac{a_{14,24}}{x_{14} x_{24}} + \frac{a_{25,35}}{x_{25} x_{35}} + \frac{a_{25,24}}{x_{25} x_{24}}$$

$$A_S(2, 3, 4, 5, 1) = \frac{a_{13,14}}{x_{24} x_{25}} + \frac{a_{13,35}}{x_{24} x_{41}} + \frac{a_{14,24}}{x_{25} x_{35}} + \frac{a_{25,35}}{x_{31} x_{41}} + \frac{a_{25,24}}{x_{31} x_{35}}$$

Conjecture: if some boly, unitarity is equivalent to every amplitude is invariant under cyclic permutations.
(thru false)

Conjecture 1: hidden zeros $\Rightarrow$ invariance under cyclic permutations.

ex: $n=5$, non-bal version

orbits: $\left[\begin{array}{c} \text{pentagon with 2 orange diagonals} \end{array} \right] \quad \left[\begin{array}{c} \text{pentagon with 2 orange diagonals} \end{array} \right] \quad \left[\begin{array}{c} \text{pentagon with 2 orange diagonals} \end{array} \right]$

$$\cancel{\text{pentagon with 2 orange diagonals}} - \text{pentagon with 2 orange diagonals} + \cancel{\text{pentagon with 2 orange diagonals}} = 0$$

$$\cancel{\text{pentagon with 2 orange diagonals}} = 0$$

$n=6$

